

Self-Service Coffee Kiosk

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1. Scenario Overview

Software is designed for a self-service coffee kiosk in a busy café. Customers can select coffee, choose size, customize add-ons, pay using card or mobile wallet, and receive a receipt with loyalty points. A single order must be completed in under 60 seconds. Admin functions support ingredient refilling, price updates, and sales reports.

2. Requirements Specification

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a customer to select a coffee type from Espresso, Americano, Latte, and Cappuccino.	High	Selecting a coffee type highlights the choice and stores it in the current order.	Core functionality required to start an order.
FR-002	Functional	The system shall allow the customer to select a drink size: Small, Medium, or Large.	High	Exactly one size can be selected and is displayed in the order summary.	Size is required for preparation and pricing.
FR-003	Functional	The system shall allow customers to customize an order with available add-ons such as an extra shot, soy/almond milk, and syrups.	Medium	Each selected add-on appears in the order summary and the corresponding price is added correctly.	Supports customer preferences.
FR-004	Functional	The system shall allow customers to place an order and pay using a credit/debit card or mobile wallet.	High	A valid payment creates a unique successful order; a failed payment does not create a completed order.	Payment is essential to complete the purchase.
FR-005	Functional	The system shall print a receipt containing order details and record loyalty points after a successful order.	Medium	The receipt contains selected items, size, add-ons, total amount, and order ID; points are added once.	Provides proof of purchase and supports loyalty.
NFR-001	Nonfunctional	The system shall allow a customer to complete a single order from selection to successful payment in under 60 seconds.	High	At least 95% of normal test orders are completed within 60 seconds.	Ensures quick service during peak hours.
NFR-002	Nonfunctional	The kiosk interface shall be simple and responsive to customer input.	High	At least 95% of valid interactions produce visible feedback within 1 second.	Improves usability and reduces customer confusion.

3. Actors and Use Cases

Actors: Customer, Payment Gateway, Admin, Receipt Printer

Use Cases:

UC-01: Select Coffee

UC-02: Choose Size

UC-03: Customize Add-ons

UC-04: Place Order

UC-05: Process Payment

UC-06: Print Receipt & Add Loyalty Points

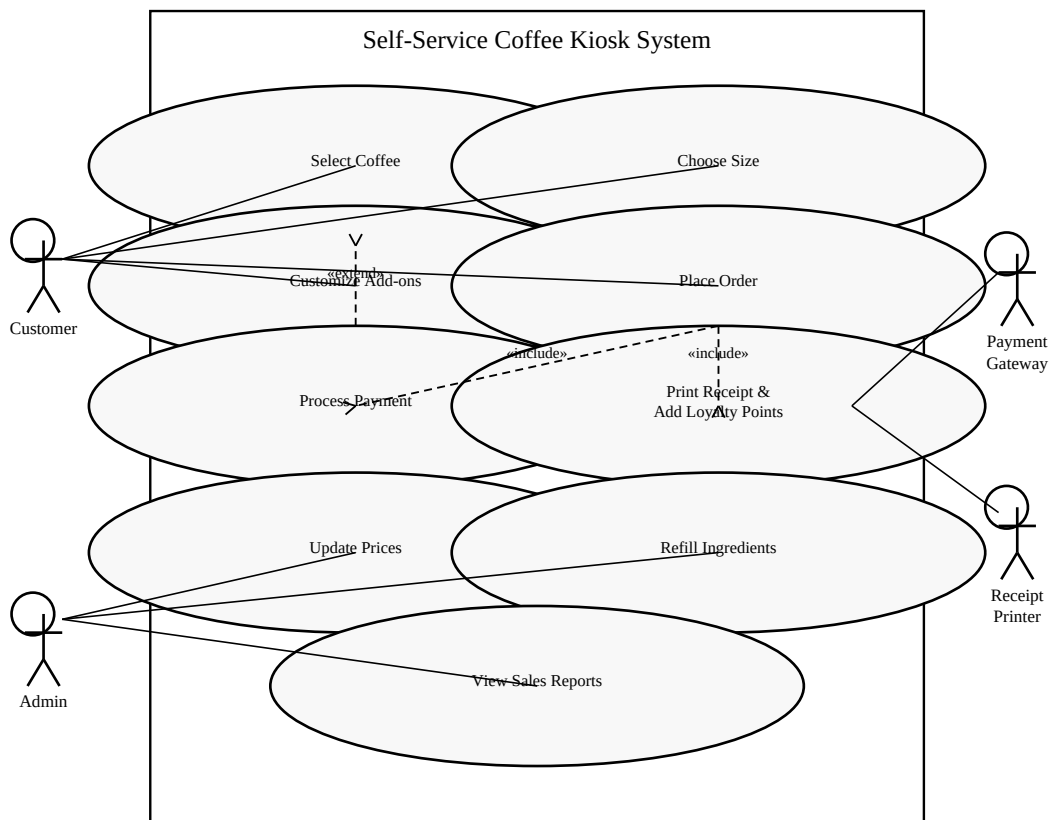
UC-07: Update Prices

UC-08: Refill Ingredients

UC-09: View Sales Reports

4. UML Use-Case Diagram

A compact UML layout showing actors outside the system boundary and use cases arranged in aligned columns. Dashed arrows indicate «include» or «extend» relationships.



5. Use-Case Flow

Primary Actor: Customer

Main Success Scenario

1. Customer starts a new order. The system displays the available coffee types.
2. Customer selects a coffee type. The system stores and highlights the selection.
3. System displays the available sizes. Customer selects Small, Medium, or Large.
4. System displays available add-ons. Customer optionally selects add-ons and the order summary is updated.
5. System displays the final order summary and total amount.
6. Customer confirms the order and selects a payment method: credit/debit card or mobile wallet.
7. System sends the payment request to the Payment Gateway.
8. Payment Gateway authorizes the payment and returns a successful response.
9. System creates the order and records the transaction.
10. System prints a receipt containing order details and adds the applicable loyalty points.
11. System displays a successful order confirmation. The use case ends successfully.

Alternate Flow – Payment Declined

- Payment Gateway returns a declined or failed response.
- The system displays a payment failure message and keeps the order available for retry.
- The customer may choose another supported payment method and return to the payment step.
- If payment fails again or the customer cancels, the order is cancelled and no completed order is created.

Pharmacy Medicine Pickup System

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1. Scenario Overview

Software is designed for a self-service medicine ordering and pickup system. Customers search and select medicines, upload prescriptions when required, place pickup orders, receive readiness notifications, collect medicines, and receive receipts. The pharmacist verifies prescriptions, checks availability, prepares orders, and manages inventory, medicines, prices, and sales reports. Orders should be placeable in under 90 seconds.

2. Requirements Specification

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow customers to search for and select required medicines.	High	Selected medicines are added to the order summary with the requested quantity.	Enables customers to create medicine orders.
FR-002	Functional	The system shall allow customers to upload or scan a doctor's prescription for prescription-only medicines.	High	A valid uploaded or scanned prescription is attached to the corresponding order.	Required for regulated medicines.
FR-003	Functional	The system shall allow customers to place a medicine pickup order after required order details are provided.	High	The system generates a unique order ID and displays order confirmation.	Creates a traceable request for processing.
FR-004	Functional	The system shall allow the pharmacist to verify prescriptions, check medicine availability, and prepare valid orders.	High	Order status changes to Ready for Pickup only after required verification and availability checks.	Ensures safe and accurate dispensing.
FR-005	Functional	The system shall notify the customer when an order is ready for pickup and provide a receipt after collection.	High	One ready notification is generated per ready order, and the receipt contains order ID, medicines, quantities, and pickup status.	Keeps the customer informed and provides proof of collection.
NFR-001	Nonfunctional	The system shall allow customers to place an order within 90 seconds.	High	At least 95% of normal test orders are completed within 90 seconds.	Reduces waiting time during busy hours.
NFR-002	Nonfunctional	The system shall provide a simple and responsive user interface.	High	At least 95% of valid user interactions receive visible feedback within 1 second under normal load.	Improves usability and efficient self-service.

3. Actors and Use Cases

Actors: Customer, Admin / Pharmacist, Notification Service, Receipt Printer

Use Cases:

UC-01: Search & Select Medicines

UC-02: Upload / Scan Prescription

UC-03: Place Medicine Order

UC-04: Verify Prescription

UC-05: Check Medicine Availability

UC-06: Prepare Order

UC-07: Notify Ready for Pickup

UC-08: Collect Medicines & Print Receipt

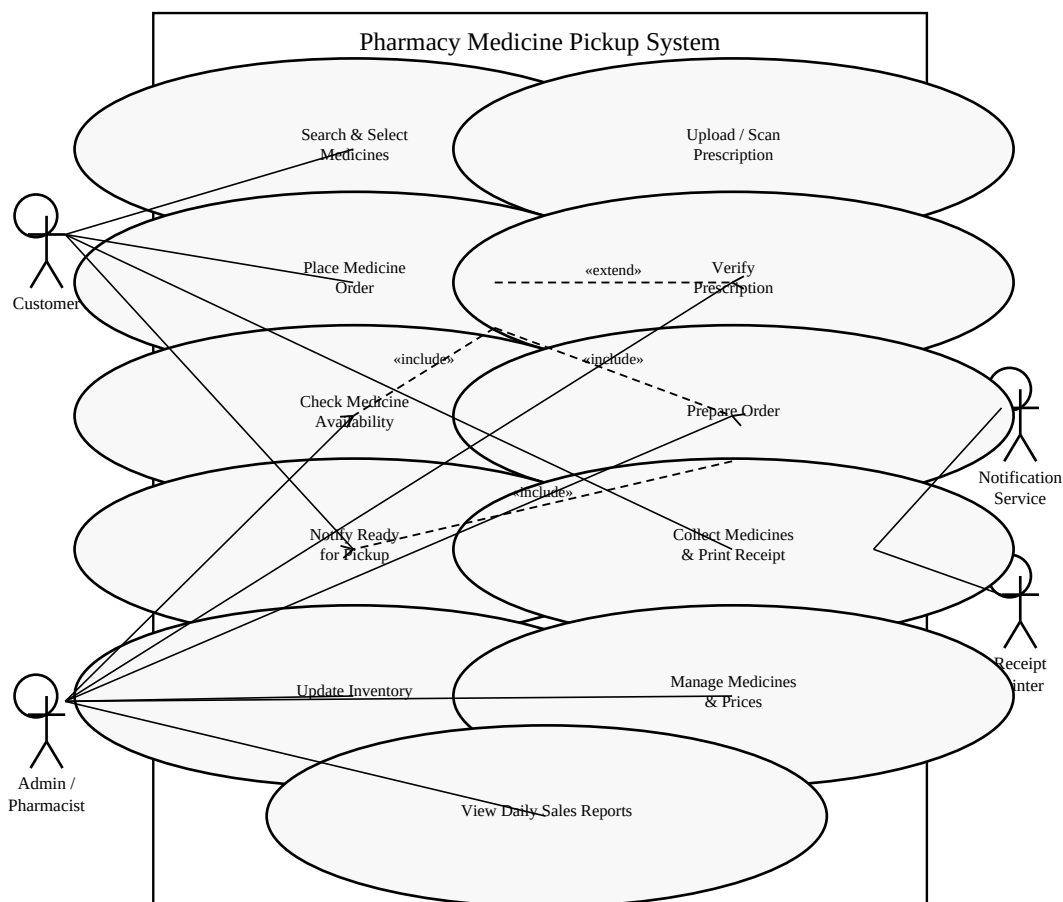
UC-09: Update Inventory

UC-10: Manage Medicines & Prices

UC-11: View Daily Sales Reports

4. UML Use-Case Diagram

A compact UML layout showing actors outside the system boundary and use cases arranged in aligned columns. Dashed arrows indicate «include» or «extend» relationships.



5. Use-Case Flow

Primary Actor: Customer

Main Success Scenario

1. Customer starts a new medicine pickup order.
2. Customer searches for and selects the required medicines and quantities.
3. If a selected medicine requires a prescription, the system requests the customer to upload or scan it.
4. Customer provides the required prescription, when applicable, and reviews the order summary.
5. Customer confirms and submits the medicine pickup order.
6. System generates a unique order ID and records the order with status Pending Review.
7. The pharmacist verifies the prescription, when applicable, and checks medicine availability.
8. The pharmacist prepares the valid and available medicines.
9. The pharmacist marks the order as Ready for Pickup.
10. The system sends a notification informing the customer that the order is ready.
11. Customer collects the medicines and receives a receipt. The use case ends successfully.

Alternate Flow – Prescription Invalid or Medicine Unavailable

- If the prescription is invalid or incomplete, the pharmacist marks the order as requiring correction and the customer is notified.
- The customer may provide a corrected prescription, after which verification is repeated.
- If a requested medicine is unavailable, the affected order may be modified or cancelled.
- An order is marked Ready for Pickup only after all required verification and preparation steps are completed.